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Date: 1/15/24

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Course: Calculus Beta

Assignment: MyLab Math - Uge 7

A single fair 20-sided die is rolled. Find the probability of getting a 19 or a 20.

Let S be a sample space of equally likely outcomes, and let event E be a subset of S . Then the probability that event E occurs is $P(E) = \frac{n(E)}{n(S)}$.

To determine $P(E)$, it is necessary to determine S , $n(S)$, E , and $n(E)$. The sample space S of possible outcomes is $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$. Determine $n(S)$.

$$n(S) = 20$$

Now, determine the event E .

$$E = \{19, 20\}$$

Now determine $n(E)$.

$$n(E) = 2$$

Substitute these values into the formula and simplify to find $P(E)$.

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(E) = \frac{2}{20}$$

$$P(E) = \frac{1}{10}$$

Thus, the probability of rolling a 19 or a 20 is $\frac{1}{10}$.